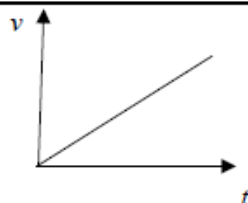


Mark Scheme

Q1.

Question	Scheme	Marks	AOs
	Integrate a w.r.t. time	M1	1.1a
	$\mathbf{v} = \frac{5t^2}{2}\mathbf{i} - 10t^{\frac{3}{2}}\mathbf{j} + \mathbf{C}$ (allow omission of C)	A1	1.1b
	$\mathbf{v} = \frac{5t^2}{2}\mathbf{i} - 10t^{\frac{3}{2}}\mathbf{j} + 20\mathbf{i}$	A1	1.1b
	When $t = 4$, $\mathbf{v} = 60\mathbf{i} - 80\mathbf{j}$	M1	1.1b
	Attempt to find magnitude: $\sqrt{(60^2 + 80^2)}$	M1	3.1a
	Speed = 100 m s^{-1}	A1ft	1.1b
(6 marks)			
Notes:			
1 st M1: for integrating a w.r.t. time (powers of t increasing by 1) 1 st A1: for a correct v expression without C 2 nd A1: for a correct v expression including C 2 nd M1: for putting $t = 4$ into their v expression 3 rd M1: for finding magnitude of their v 3 rd A1: ft for 100 m s^{-1} , follow through on an incorrect v			

Q2.

Question	Scheme	Marks	AOs
	Mark parts (a) and (b) together		
(a)	Equation of motion for A	M1	3.3
	$3mg \sin \alpha - F - T = 3ma$	A1	1.1b
		(2)	
(b)	Resolve perpendicular to the plane	M1	3.4
	$R = 3mg \cos \alpha$	A1	1.1b
	$F = \frac{1}{6}R$	B1	1.2
	Equation of motion for B OR for whole system	M1	3.3
	$T - mg = ma$ OR $3mg \sin \alpha - F - mg = 3ma + ma$	A1	1.1b
	Complete method to solve for a	DM1	3.1b
	$a = \frac{1}{10}g$ *	A1*	2.2a
		(7)	
(c)		B1	1.1b
	e.g. acceleration (of B) is constant; dependent on first B1	DB1	2.4
		(2)	
(d)	e.g. the tensions in the two equations of motion would be different. Tension on A would be different to tension on B	B1	3.5a
		(1)	
(12 marks)			

Notes: N.B. If m's are consistently missing treat as a MR, so max (a) M1A0 (b) M1A0B0M1A1M1A1 (c) B1B1 (d) B1 For (a) and (b), allow verification, but must see full equations of motion.		
a	M1	Equation in T and a with correct no. of terms, condone sign errors and sin/cos confusion (If one of the 3's is missing, allow M1) N.B. Treat sin(3/5) etc as an A error but allow recovery
	A1	Correct equation (allow $(-a)$ instead of a in <u>both</u> equations)
b	M1	Correct no. of terms, condone sign errors and sin/cos confusion Allow if appears in (a)
	A1	Correct equation
	B1	Seen anywhere in (a) or (b), including on a diagram
	M1	Equation (for B) in T and a with correct no. of terms, condone sign errors and sin/cos confusion OR Whole system equation with correct no. of terms, condone sign errors and sin/cos confusion
	A1	Correct equation
	DM1	Complete method (trig may not be substituted), dependent on M1 in (a) and second M1 in (b) if they use two equations, or second M1 in (b) if they use one equation.
	A1*	Correct answer correctly obtained.
c	B1	Straight line starting at the origin (could be reflected in the t -axis). B0 if continuous vertical line at the end.
	DB1	Dependent on first B1, for any equivalent statement
d	B1	B0 if incorrect extras

Q3.

Question	Scheme	Marks	AOs
(a)	Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$ with $t = 2$: $\mathbf{v} = 4\mathbf{i} + 2(2\mathbf{i} - 3\mathbf{j})$ OR integration: $\mathbf{v} = (2\mathbf{i} - 3\mathbf{j})t + 4\mathbf{i}$, with $t = 2$	M1	3.1a
	$\mathbf{v} = 8\mathbf{i} - 6\mathbf{j}$	A1	1.1b
		(2)	
(b)	Use of $\mathbf{r} = \mathbf{ut} + \frac{1}{2}\mathbf{at}^2$ at $t = 3$: $(\mathbf{i} + \mathbf{j}) + \left[3 \times 4\mathbf{i} + \frac{1}{2} \times (2\mathbf{i} - 3\mathbf{j}) \times 3^2 \right]$ OR: find $\mathbf{v}$ at $t = 3$: $4\mathbf{i} + 3(2\mathbf{i} - 3\mathbf{j}) = (10\mathbf{i} - 9\mathbf{j})$ then use $\mathbf{r} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$ $(\mathbf{i} + \mathbf{j}) + \left[\frac{1}{2} [4\mathbf{i} + (10\mathbf{i} - 9\mathbf{j})] \times 3 \right]$ or $\mathbf{r} = \mathbf{vt} - \frac{1}{2}\mathbf{at}^2$ $(\mathbf{i} + \mathbf{j}) + \left[3 \times (10\mathbf{i} - 9\mathbf{j}) - \frac{1}{2} \times (2\mathbf{i} - 3\mathbf{j}) \times 3^2 \right]$ OR integration: $\mathbf{r} = (\mathbf{i} + \mathbf{j}) + \left[(2\mathbf{i} - 3\mathbf{j}) \frac{1}{2}t^2 + 4\mathbf{i}t \right]$, with $t = 3$	M1	3.1a
	$\mathbf{r} = 22\mathbf{i} - 12.5\mathbf{j}$	A1	2.2a
		(2)	
(4 marks)			

Notes: Accept column vectors throughout		
a	M1	Complete method to find $\mathbf{v}$, using $\mathbf{ruvat}$ or integration (M0 if $\mathbf{i}$ and/or $\mathbf{j}$ is missing)
	A1	Apply isw if they also find the speed
b	M1	Complete method to find the p.v. but this mark can be scored if they omit $(\mathbf{i} + \mathbf{j})$ i.e. the M1 is for the expression in the square bracket If they integrate, the M1 is earned once the expression in the square bracket is seen with $t = 3$ (M0 if $\mathbf{i}$ and/or $\mathbf{j}$ is missing)
	A1	cao

Q4.

Question	Scheme	Marks	AOs
(i)(a)	Integrate a wrt t to obtain velocity	M1	3.4
	$v = (t - 2t^2)\mathbf{i} + \left(3t - \frac{1}{3}t^3\right)\mathbf{j} (+C)$	A1	1.1b
	$8\mathbf{i} - \frac{28}{3}\mathbf{j} \text{ (m s}^{-1}\text{)}$	A1	1.1b
		(3)	
(i)(b)	Equate $\mathbf{i}$ component of v to zero	M1	3.1a
	$t - 2t^2 + 36 = 0$	A1ft	1.1b
	$t = 4.5$ (ignore an incorrect second solution)	A1	1.1b
		(3)	
(ii)	Differentiate r wrt to t to obtain velocity	M1	3.4
	$v = (2t - 1)\mathbf{i} + 3\mathbf{j}$	A1	1.1b
	Use magnitude to give an equation in t only	M1	2.1
	$(2t - 1)^2 + 3^2 = 5^2$	A1	1.1b
	Solve problem by solving this equation for t	M1	3.1a
	$t = 2.5$	A1	1.1b
		(6)	
(12 marks)			

Notes: Accept column vectors throughout			
(i)(a)	M1	At least 3 terms with powers increasing by 1 (but M0 if clearly just multiplying by t)	
	A1	Correct expression	
	A1	Accept $8\mathbf{i} - 9.3\mathbf{j}$ or better. Isw if speed found.	
(i)(b)	M1	Must have an equation in t only (Must have integrated to find a velocity vector)	
	A1ft	Correct equation follow through on their v but must be a 3 term quadratic	
	A1	cao	
(ii)	M1	At least 2 terms with powers decreasing by 1 (but M0 if clearly just dividing by t)	
	A1	Correct expression	
	M1	Use magnitude to give an equation in t only, must have differentiated to find a velocity (M0 if they use $\sqrt{x^2 - y^2}$)	
	A1	Correct equation $\sqrt{(2t - 1)^2 + 3^2} = 5$	
	M1	Solve a 3 term quadratic for t which has come from differentiating and using a magnitude. This M mark can be implied by a correct answer with no working.	
	A1	2.5	

Q5.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	Horizontal speed = $20 \cos 30^\circ = 10\sqrt{3} \text{ m s}^{-1}$	B1	This mark is given for a correct expression for the horizontal speed of P
	$v = u + at$ Vertical speed = $20 \sin 30^\circ - 19.6$ $= -9.6 \text{ m s}^{-1}$	M1	This mark is given for a method to find the vertical speed of P
		A1	This mark is given for a correct value for the vertical speed of P
	$\theta = \tan^{-1} \pm \frac{9.6}{10\sqrt{3}}$	M1	This mark is given finding an expression for the value of θ
	Speed = $\sqrt{(100 \times 3) + 9.6^2}$	M1	This mark is given for using Pythagoras to find the magnitude of the speed of P
	9.8 m s^{-1} downwards at 29.0° to the horizontal	A1	This mark is given for finding the correct velocity of P (showing both magnitude and direction)
(b)	Sum of horizontal distances = 50 m	M1	This mark is given for stating the sum of the horizontal distances
	$(u \cos \theta) \times 2 = 50 - (20 \cos 30^\circ) \times 2$ $u \cos \theta = 25 - 20 \cos 30^\circ$	A1	This mark is given for a correct expression for the horizontal distance
	Vertical distances equal $(20 \sin 30^\circ) \times 2 - \frac{g}{2} \times 4 = (u \sin \theta) \times 2 - \frac{g}{2} \times 4$	M1	This mark is given for equating the vertical distances
	$u \sin \theta = 20 \sin 30^\circ$	A1	This mark is given for a correct expression for the vertical distance
	$\theta = 52.5^\circ, u = 12.6 \text{ m s}^{-1}$	M1	This mark is given for a correct method to find θ and u
		A1	This mark is given for finding correct values of θ and u
(c)	For example: The effect of the wind The effect of the spinning of the balls The size of the balls	B1	This mark is given for one correct limitation of the model stated

Q6.

Question	Scheme	Marks	AOs
	Integrate $\mathbf{v}$ w.r.t. time	M1	1.1a
	$\mathbf{r} = 2t^4\mathbf{i} - 2t^2\mathbf{j} (+ \mathbf{C})$	A1	1.1b
	Substitute $t = 4$ and $t = 1$ into their $\mathbf{r}$	M1	1.1b
	$t = 4, \mathbf{r} = 4\mathbf{i} - 32\mathbf{j} (+ \mathbf{C}); t = 1, \mathbf{r} = 2\mathbf{i} - 2\mathbf{j} (+ \mathbf{C})$ or $(4, -32); (2, -2)$	A1	1.1b
	$\sqrt{2^2 + (-30)^2}$	M1	1.1b
	$\sqrt{904} = 2\sqrt{226}$	A1	1.1b
		(6)	
(6 marks)			
Notes: Allow column vectors throughout			
M1: At least one power increasing by 1. A1: Any correct (unsimplified) expression M1: Must have attempted to integrate $\mathbf{v}$. Substitute $t = 4$ and $t = 1$ into their $\mathbf{r}$ to produce 2 vectors (or 2 points if just working with coordinates). A1: $4\mathbf{i} - 32\mathbf{j} (+ \mathbf{C})$ and $2\mathbf{i} - 2\mathbf{j} (+ \mathbf{C})$ or $(4, -32)$ and $(2, -2)$. These can be seen or implied. M1: Attempt at distance of form $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ for their points. Must have 2 non zero terms. A1: $\sqrt{904} = 2\sqrt{226}$ or any equivalent surd (exact answer needed)			

Q7.

Question	Scheme	Marks	AOs
(a)	Use of $\mathbf{r} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$: $(7\mathbf{i} - 10\mathbf{j}) = 2(2\mathbf{i} - 3\mathbf{j}) + \frac{1}{2}\mathbf{a}2^2$	M1	3.1b
	$\mathbf{a} = (1.5\mathbf{i} - 2\mathbf{j})$	A1	1.1b
	$ \mathbf{a} = \sqrt{1.5^2 + (-2)^2}$	M1	1.1b
	$= 2.5 \text{ m s}^{-2}$ * GIVEN ANSWER	A1*	2.1
		(4)	
(b)	Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t = (2\mathbf{i} - 3\mathbf{j}) + 2(1.5\mathbf{i} - 2\mathbf{j})$	M1	3.1b
	$= (5\mathbf{i} - 7\mathbf{j})$	A1	1.1b
	$\mathbf{v} = (5\mathbf{i} - 7\mathbf{j}) + t(4\mathbf{i} + 8.8\mathbf{j}) = (5 + 4t)\mathbf{i} + (8.8t - 7)\mathbf{j}$ and $(5 + 4t) = (8.8t - 7)$	M1	3.1b
	$t = 2.5 \text{ (s)}$	A1	1.1b
		(4)	
(8 marks)			

Notes: Allow column vectors throughout

(a)

No credit for individual component calculations

M1: Using a complete method to obtain the acceleration. N.B. Equation, in a only, could be obtained by two integrations

ALTERNATIVE

M1: Use velocity at half-time ($t = 1$) = Average velocity over time period

So at $t = 1$, $\mathbf{v} = \frac{1}{2}(7\mathbf{i} - 10\mathbf{j})$ so $\mathbf{a} = \frac{1}{2}(7\mathbf{i} - 10\mathbf{j}) - (2\mathbf{i} - 3\mathbf{j})$

N.B. could see $(7\mathbf{i} - 10\mathbf{j}) = (4\mathbf{i} - 6\mathbf{j}) + 2\mathbf{a}$ as first line of working

A1: Correct a vector

M1: Attempt to find magnitude of their $\mathbf{a}$ using form $\sqrt{a^2 + b^2}$

A1*: Correct GIVEN ANSWER obtained correctly

(b)

M1: Using a complete method to obtain the velocity at A e.g. by use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ with $t = 2$ and $\mathbf{u} = 2\mathbf{i} - 3\mathbf{j}$ and their $\mathbf{a}$

OR: by use of $s = \mathbf{v}t - \frac{1}{2}\mathbf{a}t^2$

OR: by integrating their $\mathbf{a}$, with addition of $\mathbf{C} = 2\mathbf{i} - 3\mathbf{j}$, and putting $t = 2$

A1: correct vector

M1: Complete method to find equation in t only

e.g. by using $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, with their $\mathbf{u}$ and equating $\mathbf{i}$ and $\mathbf{j}$ components

OR: by integrating $(4\mathbf{i} + 8.8\mathbf{j})$, with addition of a constant, and equating $\mathbf{i}$ and $\mathbf{j}$ components.

N.B. Must be equating $\mathbf{i}$ and $\mathbf{j}$ components of a velocity vector and must be their velocity at A, to give an equation in t only for this M mark

A1: 2.5 (s)

Q8.

Question	Scheme	Marks	AOs
(a)	$(4\mathbf{i} - \mathbf{j}) + (\lambda\mathbf{i} + \mu\mathbf{j}) = (4 + \lambda)\mathbf{i} + (-1 + \mu)\mathbf{j}$	M1	3.4
	Use ratios to obtain an equation in λ and μ only	M1	2.1
	$\frac{(4 + \lambda)}{(-1 + \mu)} = \frac{3}{1}$ or $\frac{\frac{1}{4}(4 + \lambda)}{\frac{1}{4}(-1 + \mu)} = \frac{3}{1}$	A1	1.1b
	$\lambda - 3\mu + 7 = 0$ * Allow $0 = \lambda - 3\mu + 7$ but nothing else.	A1*	1.1b
		(4)	

(b)		$\lambda = 2 \Rightarrow \mu = 3$; Resultant force = $(6\mathbf{i} + 2\mathbf{j})$ (N)	M1	3.1a
		$(6\mathbf{i} + 2\mathbf{j}) = 4\mathbf{a}$ OR $ (6\mathbf{i} + 2\mathbf{j}) = 4a$	M1	1.1b
		Use of $\mathbf{r} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$ with $\mathbf{u} = 0$, their $\mathbf{a}$ and $t = 4$: Or they may integrate their $\mathbf{a}$ twice with $\mathbf{u} = 0$ and put $t = 4$: $\mathbf{r} = \frac{1}{2} \times \frac{(6\mathbf{i} + 2\mathbf{j})}{4} 4^2 = (12\mathbf{i} + 4\mathbf{j})$	DM1	2.1
		$\sqrt{12^2 + 4^2}$	M1	1.1b
		ALTERNATIVE 1 for last two M marks: Use of $s = ut + \frac{1}{2}at^2$, with $u = 0$, their a and $t = 4$: DM1 $s = \frac{1}{2} \times \sqrt{1.5^2 + 0.5^2} \times 4^2$ Use of Pythagoras to find mag of $\mathbf{a}$: $a = \sqrt{1.5^2 + 0.5^2}$ M1		
		ALTERNATIVE 2 for last two M marks: Use of $s = ut + \frac{1}{2}at^2$, with $u = 0$, their a and $t = 4$: DM1 $s = \frac{1}{2} \times \left(\frac{\sqrt{6^2 + 2^2}}{4} \right) \times 4^2$ Use of Pythagoras to find $ (6\mathbf{i} + 2\mathbf{j}) $: $= \sqrt{6^2 + 2^2}$ M1		
		$\sqrt{160}$, $2\sqrt{40}$, $4\sqrt{10}$ oe or 13 or better (m)	A1	1.1b
			(5)	
(9 marks)				
Notes: Accept column vectors throughout				
a	M1	Adding the two forces, $\mathbf{i}$'s and $\mathbf{j}$'s must be collected (or must be a single column vector) seen or implied		
	M1	Must be using ratios; Ignore an equation e.g. $(4 + \lambda)\mathbf{i} + (-1 + \mu)\mathbf{j} = 3\mathbf{i} + \mathbf{j}$ if they go on to use ratios.		

		However, if they write $4 + \lambda = 3$ and $-1 + \mu = 1$ then $3(-1 + \mu) = 3$ so $4 + \lambda = 3(-1 + \mu)$ with no use of a constant, it's M0 They may use the acceleration, with a factor of $\frac{1}{4}$ top and bottom, see alternative Allow one side of the equation to be inverted
	A1	Correct equation
	A1*	Given answer correctly obtained. Must see at least one line of working, with the LH fraction 'removed'.
b	M1	Adding F_1 and F_2 to find the resultant force, λ and μ must be substituted N.B. M0 if they use $\mu = 2$ coming from $-1 + \mu = 1$ in part (a).
	M1	Use of $F = 4a$ Or $F = 4a$, where F is <u>their</u> resultant. (including $3i + j$) This is an independent mark, so could be earned, for example, if they have subtracted the forces to find the 'resultant' N.B. M0 if only using F_1 or F_2
	DM 1	Dependent on previous M mark for Either: use of $r = ut + \frac{1}{2}at^2$ with $u = 0$, their a and $t = 4$ to produce a displacement vector Or : integrate twice, with $u = 0$, their a and $t = 4$ to produce a displacement Vector Or: use of $s = ut + \frac{1}{2}at^2$ with $u = 0$, their a and $t = 4$ to produce a length
	M1	Use of Pythagoras, with square root, to find the magnitude of their displacement vector, a or F (M0 if only using F_1 or F_2) depending on which method they have used.
	A1	cao

Q9.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\mathbf{v} = \mathbf{u} + \mathbf{at}$ $\mathbf{v} = (-\mathbf{i} + 4\mathbf{j}) + (2\mathbf{i} - 3\mathbf{j})t$	M1	This mark is given for a method to find a vector expression for $\mathbf{v}$
	$= (-1 + 2t)\mathbf{i} + (4 - 3t)\mathbf{j}$	A1	This mark is given for finding a correct vector expression for $\mathbf{v}$
	$\frac{4 - 3T}{1 + 2T} = \frac{-4}{3}$	M1	This mark is given for a correct use of ratios as a method to find the value of T
	$12 - 9T = 4 - 8T$ $T = 12 - 4 = 8$	A1	This mark is given for finding the correct value of T
(b)	$\mathbf{s} = \mathbf{ut} + \frac{1}{2}\mathbf{at}^2$ $\mathbf{s} = (-\mathbf{i} + 4\mathbf{j})t + \frac{1}{2}(2\mathbf{i} - 3\mathbf{j})t^2$	M1	This mark is given for a method to find a vector expression for the distance AB
	$= (-t + t^2)\mathbf{i} + \left(4t - \frac{3}{2}t^2\right)\mathbf{j}$	A1	This mark is given for finding a correct vector expression for the distance AB
	$AB = \sqrt{12^2 + 8^2}$	M1	This mark is given for a method to find the distance AB using Pythagoras and substituting $t = 4$
	$= 14.4 \text{ m}$	A1	This mark is given for find a correct value for the distance AB
(Total 8 marks)			

Q10.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\mathbf{a} = 6\mathbf{i} - \frac{15}{2}t^{\frac{1}{2}}\mathbf{j}$	M1	This mark is given for a method to differentiate the expression for $\mathbf{v}$
		A1	This mark is given for correctly differentiating the expression for $\mathbf{v}$
	$= 6\mathbf{i} - 15\mathbf{j} \text{ m s}^{-1}$	A1	This mark is given for substituting $t = 4$ to find a correct vector expression for the acceleration of P
(b)	$\mathbf{r} = (\mathbf{r}_0) + 3t^2\mathbf{i} - 2t^{\frac{5}{2}}\mathbf{j}$	M1	This mark is given for a method to integrate the expression for $\mathbf{v}$
		A1	This mark is given for correctly integrating the expression for $\mathbf{v}$
	$(-20\mathbf{i} + 20\mathbf{j}) + (48\mathbf{i} - 64\mathbf{j})$ $= 28\mathbf{i} - 44\mathbf{j} \text{ m}$	A1	This mark is given for substituting $t = 4$ to find a correct position vector of P
(Total 6 marks)			

Q11.

Question	Scheme	Marks	AOs
	Allow column vectors throughout this question		
(a)	Differentiate $\mathbf{v}$ wrt t	M1	3.1a
	$\frac{3}{2}t^{-\frac{1}{2}}\mathbf{i} - 2\mathbf{j}$ isw	A1	1.1b
		(2)	
(b)	$3t^{\frac{1}{2}} = 2t$	M1	2.1
	Solve for t	DM1	1.1b
	$t = \frac{9}{4}$	A1	1.1b
		(3)	
(c)	Integrate $\mathbf{v}$ wrt t	M1	3.1a
	$\mathbf{r} = 2t^{\frac{3}{2}}\mathbf{i} - t^2\mathbf{j} (+C)$	A1	1.1b
	$t = 1, \mathbf{r} = -\mathbf{j} \Rightarrow C = -2\mathbf{i}$ so $\mathbf{r} = 2t^{\frac{3}{2}}\mathbf{i} - t^2\mathbf{j} - 2\mathbf{i}$	A1	2.2a
		(3)	
(d)	$\sqrt{(3t^{\frac{1}{2}})^2 + (2t)^2} = 10$ or $(3t^{\frac{1}{2}})^2 + (2t)^2 = 10^2$	M1	2.1
	$9t + 4t^2 = 100$	M(A)1	1.1b
	$t = 4$	A1	1.1b
	$\mathbf{r} = 14\mathbf{i} - 16\mathbf{j}$	M1	1.1b
	$\sqrt{14^2 + (-16)^2}$	M1	3.1a
	$\sqrt{452} \ (2\sqrt{113}) \text{ (m)}$	A1	1.1b
		(6)	
(14 marks)			

Notes:		
a	M1	Both powers decreasing by 1 (M0 if vector(s) disappear but allow recovery)
	A1	cao
b	M1	Complete method, using v , to obtain an equation in t only, allow a sign error
	DM1	Dependent on M1, solve for t
	A1	cao
c	M1	Both powers increasing by 1 (M0 if vectors disappear but allow recovery)
	A1	Correct expression without C
	A1	cao
d	M1	Use of Pythagoras on v and 10 to set up equation in t
	M(A)1	Correct 3 term quadratic in t
	A1	cao
	M1	Substitute their numerical t value into their r
	M1	Use of Pythagoras to find the magnitude of their r
	A1	cso

Question	Scheme		Marks	AOs
(a)	Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ or integrate to give: $\mathbf{v} = (-2\mathbf{i} + 2\mathbf{j}) + 2(4\mathbf{i} - 5\mathbf{j})$		M1	3.1a
	$(6\mathbf{i} - 8\mathbf{j}) (\text{m s}^{-1})$		A1	1.1b
			(2)	
(b)	Solve problem through use of $\mathbf{r} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$ or integration (M0 if $\mathbf{u} = 0$) Or any other complete method e.g use $\mathbf{v} = \mathbf{u} + \mathbf{a}T$ and $\mathbf{r} = \frac{(\mathbf{u} + \mathbf{v})T}{2}$:		M1	3.1a
	$-4.5\mathbf{j} = 2t\mathbf{j} - \frac{1}{2}t^2 5\mathbf{j}$ (j terms only)		A1	1.1b
	The first two marks could be implied if they go straight to an algebraic equation.			
	Attempt to equate j components to give equation in T only $(-4.5 = 2T - \frac{5}{2}T^2)$		M1	2.1
	$T = 1.8$		A1	1.1b
			(4)	
(c)	Solve problem by substituting <u>their</u> T value (M0 if $T < 0$) into the i component equation to give an equation in λ only: $\lambda = -2T + \frac{1}{2}T^2 \times 4$		M1	3.1a
	$\lambda = 2.9$ or 2.88 or $\frac{72}{25}$ oe		A1	1.1b
			(2)	
Notes: Accept column vectors throughout (8 marks)				
Notes: Accept column vectors throughout (8 marks)				
2a	M1	For any complete method to give a $\mathbf{v}$ expression with correct no. of terms with $t = 2$ used, so if integrating, must see the initial velocity as the constant. Allow sign errors.		
	A1	cao isw if they go on to find the speed.		
2b	M1	For any complete method to give a vector expression for j component of displacement in t (or T) only, using $\mathbf{a} = (4\mathbf{i} - 5\mathbf{j})$, so if integrating, RHS of equation must have the correct structure. Allow sign errors.		
	A1	Correct j vector equation in t or T . Ignore i terms.		
	M1	Must have earned 1 st M mark. Equate j components to give equation in T (allow t) only (no j's) which has come from a displacement. Equation must be a 3 term quadratic in T .		
	A1	cao		
2c	M1	Must have earned 1 st M mark in (b) Complete method - must have an equation in λ only (no i's) which has come from an appropriate displacement.. (e.g M0 if $\mathbf{a} = 0$ has been used) Expression for λ must be a quadratic in T		
	A1	cao		

Q13.

Question	Scheme	Marks	AOs
(a)	Using horizontal motion	M1	3.3
	$U \cos 45^\circ t = 100$	A1	1.1b
	Using vertical motion	M1	3.4
	$U \sin 45^\circ t - \frac{1}{2}gt^2 = -25$	A1	1.1b
	Solve problem by eliminating t and solving for U	M1	3.1b
	$U = 28^*$	A1*	1.1b
		(6)	
(b)	Using vertical motion	M1	3.4
	$0^2 = (28 \sin 45^\circ)^2 - 2gh$	A1	1.1b
	Greatest height = 45 m	A1	1.1b
		(3)	
(c)	New value > 28	B1	3.5a
		(1)	
(d)	e.g. wind effects, more accurate value of g , spin of ball, include size of the ball, not model as a particle, shape of ball	B1	3.5c
		(1)	
(11 marks)			

Notes:		
a	M1	Complete method to give equation in U and t only, condone sin/cos confusion and sign errors
	A1	Correct equation
	M1	Complete method to give equation in U and t only, condone sin/cos confusion and sign errors
	A1	Correct equation (g does not need to be substituted)
	M1	Must have earned the previous two M marks. Eliminate t and solve for U . N.B. They may solve for t first ($100 - \frac{1}{2}gt^2 = -25$) and then use it to find U .
	A1*	Exact given answer correctly obtained with no wrong working (e.g. $g = 9.81$ used) or approximation seen.
b	M1	Complete method to give equation in h only (allow if U not substituted), condone sin/cos confusion and sign errors
	A1	Correct equation (g does not need to be substituted) (A0 if U is used instead of 28)
	A1	cao
c	B1	Clear statement
d	B1	Penalise incorrect extras i.e. B0 if there are incorrect extras.
		The ground being horizontal, the cliff being vertical, ... are not part of the model so B0 Include weight/mass of the ball B0

Q14.

Question	Scheme	Marks	AOs
	Note that $g = 10$; penalise once for whole question if $g = 9.8$		
(a)	Use $s = ut + \frac{1}{2}at^2$ vertically or any complete method to give an equation in t only	M1	3.4
	$-70 = 65 \sin \alpha \times t - \frac{1}{2} \times g \times t^2$	A1	1.1b
		M(A)1	1.1b
	$t = 7$ (s)	A1	1.1b
		(4)	
(b)	Horizontal velocity component at $A = 65 \cos \alpha$ (60)	B1	3.4
	Complete method to find vertical velocity component at A	M1	3.4
	$65 \sin \alpha - g \times 7$ OR $\sqrt{(-25)^2 + 2g \times 70}$ (45)	A1ft	1.1b
	Sub for trig and square, add and square root : $\sqrt{60^2 + (-45)^2}$	M1	3.1b
	75 Accept 80 (m s^{-1})	A1	1.1b
		(5)	
(c)	e.g. an approximate value of g has been used, the dimensions of the stone could affect its motion, spin of the stone, $g = 10$ instead of 9.8 has been used, g has been assumed to be constant, wind effect, shape of the stone	B1	3.5b
		(1)	
(10 marks)			
Notes:			
a	M1	Complete method, correct no. of terms, condone sign errors and sin/cos confusion	
	A1	Correct equation in t only with at most one error	
	M(A)1	Correct equation in t only	
		N.B. For 'up and down' methods etc, the two A marks are for all the equations that they use, lose a mark for each error.	
	A1	Cao ($g = 9.8$, 7.1 or 7.11) ($g = 9.81$, 7.1 or 7.12)	
b	B1	Seen, including on a diagram.	
	M1	Condone sign errors and sin/cos confusion	
	A1ft	Correct expression; accept negative of this, follow their t	
	M1	Sub for trig and use Pythagoras	
	A1	Cao ($g = 9.8$ or 9.81, 75 or 74.8)	
c	B1	B0 if incorrect extras	

Q15.

Question	Scheme	Marks	AOs
(a)	Using the model and horizontal motion: $s = ut$	M1	3.4
	$36 = U t \cos \alpha$	A1	1.1b
	Using the model and vertical motion: $s = ut + \frac{1}{2}at^2$	M1	3.4
	$-18 = U t \sin \alpha - \frac{1}{2}gt^2$	A1	1.1b
	Correct strategy for solving the problem by setting up two equations in t and U and solving for U	M1	3.1b
	$U = 15$	A1	1.1b
	(6)		
(b)	Using the model and horizontal motion: $U \cos \alpha$ (12)	B1	3.4
	Using the model and vertical motion: $v^2 = (U \sin \alpha)^2 + 2(-10)(-7.2)$	M1	3.4
	$v = 15$	A1	1.1b
	Correct strategy for solving the problem by finding the horizontal and vertical components of velocity and combining using Pythagoras: Speed = $\sqrt{(12^2 + 15^2)}$	M1	3.1b
	$\sqrt{369} = 19 \text{ m s}^{-1}$ (2sf)	A1 ft	1.1b
	(5)		
(c)	Possible improvement (see below in notes)	B1	3.5c
	Possible improvement (see below in notes)	B1	3.5c
	(2)		
(13 marks)			

Notes:
(a) 1st M1: for use of $s = ut$ horizontally 1st A1: for a correct equation 2nd M1: for use of $s = ut + \frac{1}{2}at^2$ vertically 2nd A1: for a correct equation 3rd M1: for correct strategy (need both equations) 2nd A1: for $U = 15$
(b) B1: for $U\cos\alpha$ used as horizontal velocity component 1st M1: for attempt to find vertical component 1st A1: for 15 2nd M1: for correct strategy (need both components) 2nd A1ft: for 19 m s^{-1} (2sf) following through on incorrect component(s)
(c) B1, B1: for any two of - e.g. Include air resistance in the model of the motion - e.g. Use a more accurate value for g in the model of the motion - e.g. Include wind effects in the model of the motion - e.g. Include the dimensions of the stone in the model of the motion

Q16.

Question	Scheme	Marks	AOs
(a)	Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$: $(10.5\mathbf{i} - 0.9\mathbf{j}) = 0.6\mathbf{j} + 15\mathbf{a}$	M1	3.1b
	$\mathbf{a} = (0.7\mathbf{i} - 0.1\mathbf{j}) \text{ m s}^{-2}$ Given answer	A1	1.1b
		(2)	
(b)	Use of $\mathbf{r} = \mathbf{ut} + \frac{1}{2} \mathbf{at}^2$	M1	3.1b
	$\mathbf{r} = 0.6\mathbf{j} t + \frac{1}{2} (0.7\mathbf{i} - 0.1\mathbf{j}) t^2$	A1	1.1b
		(2)	
(c)	Equating the $\mathbf{i}$ and $\mathbf{j}$ components of $\mathbf{r}$	M1	3.1b
	$\frac{1}{2} \leftarrow 0.7 t^2 = 0.6 t - \frac{1}{2} \leftarrow 0.1 t^2$	A1ft	1.1b
	$t = 1.5$	A1	1.1b
		(3)	
(d)	Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$: $\mathbf{v} = 0.6\mathbf{j} + (0.7\mathbf{i} - 0.1\mathbf{j}) t$	M1	3.1b
	Equating the $\mathbf{i}$ and $\mathbf{j}$ components of $\mathbf{v}$	M1	3.1b
	$t = 0.75$	A1 ft	1.1b
		(3)	
(10 marks)			

Notes:
(a) M1: for use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$ A1: for given answer correctly obtained
(b) M1: for use of $\mathbf{r} = \mathbf{ut} + \frac{1}{2} \mathbf{at}^2$ A1: for a correct expression for $\mathbf{r}$ in terms of t
(c) M1: for equating the $\mathbf{i}$ and $\mathbf{j}$ components of their $\mathbf{r}$ A1ft: for a correct equation following their $\mathbf{r}$ A1: for $t = 1.5$
(d) M1: for use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$ for a general t M1: for equating the $\mathbf{i}$ and $\mathbf{j}$ components of their $\mathbf{v}$ A1ft: for $t = 0.75$, or a correct follow through answer from an incorrect equation

Q17.

Question	Scheme	Marks	AOs
(a)	Using horizontal motion	M1	3.3
	Whole Motion Half way		
	$U \cos \alpha \times t = 120$ $U \cos \alpha \times t = 60$	A1	1.1b
	Using vertical motion OR	M1	3.4
	$U \sin \alpha \times t - \frac{1}{2}gt^2 = 0$ $0 = U \sin \alpha - gt$	A1	1.1b
	Attempt to solve problem by eliminating t	DM1	3.1b
	$U^2 \sin \alpha \cos \alpha = 588^*$	A1*	2.2a
		(6)	
	N.B. No credit given if they use the given answer from (b).		

(b)	Using vertical motion OR conservation of energy	M1	3.4
	$0^2 = (U \sin \alpha)^2 - 2g \times 10$ $\frac{1}{2}mU^2 - \frac{1}{2}m(U \cos \alpha)^2 = mg \times 10$	A1	1.1b
	ALTERNATIVE 1: If t is time to top: use of $10 = \frac{1}{2}gt^2$ or $(t = \frac{10}{7})$ to obtain an equation in U and α only M1 $U \sin \alpha = 14$ or $U \cos \alpha = 42$ A1		
	ALTERNATIVE 2: If t is time to top: use of: $10 = U \sin \alpha t - \frac{1}{2}gt^2$ with $t = \frac{60}{U \cos \alpha}$ substituted to obtain an equation in U and α only: M1 $10 = U \sin \alpha \times \frac{60}{U \cos \alpha} - \frac{1}{2}g \left(\frac{60}{U \cos \alpha} \right)^2$ A1		
	Attempt to solve problem by eliminating α : e.g. $U \sin \alpha = 14 \Rightarrow U \cos \alpha = 42$, from part (a) or from using $t = \frac{10}{7}$, then square and add to give result OR: $U^2 \sin^2 \alpha = 20g = 196$ and $U^2 \sin \alpha \cos \alpha = 588$, divide to give $\tan \alpha = \frac{1}{3}$ then $\sin^2 \alpha = \frac{1}{10}$, hence result OR in ALTERNATIVE 2: sub for U^2 using part (a), to give $\tan \alpha = \frac{1}{3}$ then $\sin^2 \alpha = \frac{1}{10}$, hence result	DM1	3.1b

Q18.

	N.B. Just stating that $\sin^2 \alpha = \frac{1}{10}$, with no working is DM0A0.		
	$U^2 = 1960$ *	A1*	2.2a
	N.B. Verification (i.e. starting with $U^2 = 1960$ and trying to work backwards) is not an acceptable method for this question.		
		(4)	
(c)	V , since air resistance has to be overcome, or just 'because of <u>air resistance</u> ' isw	B1	3.5a
		(1)	
(d)	e.g. wind effects, more accurate value of g , spin of ball, size of ball, shape of ball, dimensions of ball, not a particle, variable acceleration, surface area of ball, humidity. Allow wind resistance and rotational resistance (Ignore any mention of air resistance or drag)	B1	3.5c
		(1)	
(12 marks)			

Notes:		
a		N.B. Could score 2/6 for any one of the 4 given equations if there is no corresponding second equation or there is an attempt but it's incorrect.
	M1	Complete method to give equation in U , α and t only, condone sin/cos confusion and sign errors, each term that needs to be resolved must be resolved
	A1	Correct equation
	M1	Complete method to give equation in U , α and t only, condone sin/cos confusion and sign errors, each term that needs to be resolved must be resolved
	A1	Correct equation
	DM1	Eliminate t , dependent on first and second M1's
	A1*	Given answer correctly obtained, <u>with no wrong working seen</u> . Allow $588 = U^2 \sin \alpha \cos \alpha$ but nothing else
b	M1	Complete method to give equation in U and α only with correct no. of terms, condone sin/cos confusion and sign errors, each term that needs to be resolved must be resolved
	A1	Correct equation
	DM1	Eliminate α and rearrange, dependent on first M1
	A1*	Given answer correctly obtained with <u>no wrong working seen</u> (N.B. If they use a value for α (18.43°) they lose the final A1*)
c	B1	Clear statement isw
d	B1	B0 if there is an incorrect extra e.g. mass or weight

Question	Scheme	Marks	AOs
(a)	Using the model and vertical motion: $0^2 = (U \sin \alpha)^2 - 2g \leftrightarrow (3-2)$	M1	3.3
	$U^2 = \frac{2g}{\sin^2 \alpha}$ * GIVEN ANSWER	A1*	2.2a
		(2)	
(b)	Using the model and horizontal motion: $s = ut$	M1	3.4
	$20 = Ut \cos \alpha$	A1	1.1b
	Using the model and vertical motion: $s = ut + \frac{1}{2}at^2$	M1	3.4
	$-\frac{5}{4} = Ut \sin \alpha - \frac{1}{2}gt^2$	A1	1.1b
	sub for t : $-\frac{5}{4} = U \sin \alpha \left(\frac{20}{U \cos \alpha} \right) - \frac{1}{2}g \left(\frac{20}{U \cos \alpha} \right)^2$	M1 (I)	3.1b
	sub for U^2	M1(II)	3.1b
	$-\frac{5}{4} = 20 \tan \alpha - 100 \tan^2 \alpha$	A1(I)	1.1b
	$(4 \tan \alpha - 1)(100 \tan \alpha + 5) = 0$	M1(III)	1.1b
	$\tan \alpha = \frac{1}{4} \Rightarrow \alpha = 14^\circ$ or better	A1(II)	2.2a
		(9)	
	N.B. For the last 5 marks, they may set up a quadratic in t , by substituting for $U \sin \alpha$ first, then solve the quadratic to find the value of t , then use $20 = Ut \cos \alpha$ to find α . The marks are the same but earned in a different order. Enter on ePen in the corresponding M and A boxes above, as indicated below.		

	Sub for $U \sin \alpha$ to give equation in t only	M1(II)	
	$-\frac{5}{4} = \sqrt{2g}t - \frac{1}{2}gt^2$	A1(I)	
	Solve for t	M1(III)	
	$t = \frac{5}{\sqrt{2g}}$ or 1.1 or 1.13 and use $20 = Ut \cos \alpha$	M1(I)	
	$\alpha = 14^\circ$ or better	A1(II)	
(b)	ALTERNATIVE		
	Using the model and horizontal motion: $s = ut$	M1	3.4
	$20 = Ut \cos \alpha$	A1	1.1b
	A to top: $s = vt - \frac{1}{2}at^2$ <u>and</u> top to T: $s = ut + \frac{1}{2}at^2$		
	$1 = \frac{1}{2}gt_1^2 \Rightarrow t_1 = \sqrt{\frac{2}{g}}$ <u>and</u> $\frac{9}{4} = \frac{1}{2}gt_2^2 \Rightarrow t_2 = \frac{3}{\sqrt{2g}}$ Total time $t = t_1 + t_2$	M1	3.4
	$= \sqrt{\frac{2}{g}} + \frac{3}{\sqrt{2g}} \quad (= \frac{5}{\sqrt{2g}})$	A1	1.1b
	$20 = U \frac{5}{\sqrt{2g}} \cos \alpha$ (sub. for t)	M1	3.1b
	$20 = \sqrt{\frac{2g}{\sin^2 \alpha}} \frac{5}{\sqrt{2g}} \cos \alpha$ (sub. for U)	M1	3.1b
	$\tan \alpha = \frac{1}{4}$	A1	1.1b
	Solve for α	M1	1.1b
	$\Rightarrow \alpha = 14^\circ$ or better	A1	2.2a
		(9)	

(c)	The target will have dimensions so in practice there would be a range of possible values of α Or There will be air resistance Or The ball will have dimensions Or Wind effects Or Spin of the ball	B1	3.5b
		(1)	
(d)	Find U using their α e.g. $U = \sqrt{\frac{2g}{\sin^2 \alpha}}$	M1	3.1b
	Use $20 = Ut \cos \alpha$ (or use vertical motion equation)	A1 M1	1.1b
	$t = \frac{5}{\sqrt{2g}}$ or 1.1 or 1.13	B1 A1	1.1b
		(3)	
(d)	ALTERNATIVE		
	A to top: $s = vt - \frac{1}{2}at^2$ and top to T : $s = ut + \frac{1}{2}at^2$	M1	3.1b
	$1 = \frac{1}{2}gt_1^2 \Rightarrow t_1 = \sqrt{\frac{2}{g}}$ and $\frac{9}{4} = \frac{1}{2}gt_2^2 \Rightarrow t_2 = \frac{3}{\sqrt{2g}}$ Total time $t = t_1 + t_2$	A1 M1	1.1b
	$= \sqrt{\frac{2}{g}} + \frac{3}{\sqrt{2g}} (= \frac{5}{\sqrt{2g}}) = 1.1 \text{ or } 1.13 \text{ (s)}$	B1 A1	1.1b
		(3)	
(15 marks)			



Notes:

(a)

M1: Or any other complete method to obtain an equation in U , g and α only

A1*: Correct GIVEN ANSWER

(b)

M1: Using horizontal motion

A1: Correct equation

M1: Using vertical motion . N.B. M0 if they use $s = \pm 2$ or ± 3 , but allow $s = \pm 1.25$ or ± 0.75 or ± 2.25 or ± 2.75

A1: Correct equation

M1: Using $20 = Ut \cos \alpha$ to sub. for t M1: Substituting for U^2 using (a)A1: Correct quadratic equation (in $\tan \alpha$ or $\cot \alpha$)M1: Solve a 3 term quadratic, either by factorisation or formula (or by calculator (implied) if answer is correct) and find α A1: $\alpha = 14^\circ$ or better (No restriction on accuracy since g 's cancel)

N.B. If answer is correct, previous M mark can be implied, but if answer is incorrect, an explicit attempt to solve must be seen to earn the previous M mark.

(b) ALTERNATIVE

M1: Using the model with the usual rules applying to the equation

A1: Correct equation

M1: Using the model to obtain the total time from A to T A1: Correct total time t M1: Substitute for t in $20 = Ut \cos \alpha$ M1: Substitute for U in $20 = Ut \cos \alpha$, using part (a)A1: Correct equation in $\tan \alpha$ onlyM1: Solve equation for α A1: $\alpha = 14^\circ$ or better (No restriction on accuracy since g 's cancel)

N.B. If they quote the equation of the trajectory $y = x \tan \alpha - \frac{gx^2}{2U^2 \cos^2 \alpha}$ or AND put in values for x and y , could score first 5 marks, M1A1M1A1M1 (nothing for the equation only); wrong x value loses first A mark and wrong y value loses second A mark

(c)

B1: Give one limitation of the model e.g. the ball will have dimensions, or there will be air resistance or wind effects or spin

N.B. B0 if any incorrect extra(s) but ignore extra consequences.

(d)

M1: Using their α to find a value for U A1: Treat as M1: Using their U to find a value for t B1: Treat as A1: $t = 1.1$ or 1.10 (since depends on $g = 9.8$)

(d) ALTERNATIVE

M1: Using their α to find a value for U A1: Treat as M1: Using their U to find a value for t B1: Treat as A1: $t = 1.1$ or 1.10 (since depends on $g = 9.8$)

Q19.

Question	Scheme	Marks	AOs
(a)	Put $t = 2$ in v and use Pythagoras: $\sqrt{12^2 + (-6\sqrt{2})^2}$	M1	3.1a
	$\sqrt{216}, 6\sqrt{6}$ or 15 or better (m s^{-1})	A1	1.1b
		(2)	
(b)	Differentiate v wrt t to obtain a	M1	3.4
	$6\mathbf{i} - 3t^{-\frac{1}{2}}\mathbf{j}$ or (m s^{-2}) isw	A1	1.1b
		(2)	
(c)	Integrate v wrt t to obtain r	M1	3.4
	$\mathbf{r} = t^3\mathbf{i} - 4t^{\frac{3}{2}}\mathbf{j} (+C)$	A1	1.1b
	$(\mathbf{i} - 4\mathbf{j}) = 4^3\mathbf{i} - 4 \times 4^{\frac{3}{2}}\mathbf{j} + C$	M1	3.1a
	$(-62\mathbf{i} + 24\mathbf{j}) (\text{m})$ isw e.g. if they go on to find the distance.	A1	1.1b
		(4)	
(8 marks)			

Notes: Accept column vectors throughout apart from the answer to (b).

a	M1	Need square root but -ve sign not required. Allow $\mathbf{i}$'s and/or $\mathbf{j}$'s to go missing from their v at $t = 2$, provided they have applied Pythagoras correctly.
	A1	cao N.B. Correct answer with no working can score 2 marks.
b	M1	Both powers decreasing by 1. Allow a column vector. M0 if $\mathbf{i}$ or $\mathbf{j}$ is missing but allow recovery in (b).
	A1	cao. Do not accept a column vector.
c	M1	Both powers increasing by 1 M0 if $\mathbf{i}$ or $\mathbf{j}$ is missing but allow recovery.
	A1	$(\mathbf{r} =)$ not required
	M1	Putting $\mathbf{r} = (\mathbf{i} - 4\mathbf{j})$ and $t = 4$ into their displacement vector expression which must have C (allow C) to give an equation in C only, seen or implied. Must have attempted to integrate v for this mark to be available. N.B. C does not need to be found and <u>this is a method mark, so allow slips.</u>
	A1	cao